AGH University of Science and Technology named after Stanisław Staszic in Kraków

REPORT

for the course:

Industrial Process Identification

Project topic:

Stabilization of a Pendulum on a Cart

Made by:

Jan Sosnowiec, Norbert Boroń

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1 Introduction

The objective of the project was the identification and design of a controller for a system with unknown parameters in the form of a physical pendulum mounted on a cart driven by a DC motor. Ultimately, the controller should enable maintaining the pendulum in the upright position.

2 Mathematical model

The system was modeled using a system of differential equations with four state variables:

$$
\begin{aligned}
\dot{x}_1 &= x_3, \\
\dot{x}_2 &= x_4, \\
\dot{x}_3 &= \frac{a(u - \mu x_4^2 \sin x_2 - f_c x_3) + l \cos x_2 (\mu g \sin x_2 - f_p x_4)}{J_0 - \mu l \cos^2 x_2}, \\
\dot{x}_4 &= \frac{l \cos x_2 (u - \mu x_4^2 \sin x_2 - f_c x_3) + \mu g \sin x_2 - f_p x_4}{J_0 - \mu l \cos^2 x_2},
\end{aligned}
\tag{1}
$$

gdzie:
x_1 - cart position $[m]$,
x_2 - pendulum deflection $[rad]$,
x_3 - cart velocity $[m/s]$,
x_4 - pendulum angular velocity $[rad/s]$,
u - force applied by the motor to the cart (control input) $[N]$,

m_c - mass of the cart $[kg]$,
m_p - mass of the pendulum $[kg]$,
f_p - friction coefficient in the pendulum bearing $[kg \cdot m^2/s]$,
f_c - friction coefficient in the cart bearings $[kg/s]$,
l - distance from the system's center of mass to the pendulum axis $[m]$,
J_0 - moment of inertia of the pendulum $[kg \cdot m^2]$,
$g = 9.81 m/s^2$ - gravitational acceleration,
$\mu = l(m_c + m_p)$,
$a = l^2 + \frac{J_0}{m_c + m_p}$.

Additionally, the motor controlling the system was approximated by a linear model:

$$F = kU, \tag{2}$$

where:
F - force exerted by the motor on the cart $[N]$,
U - control voltage $[V]$,
k - coefficient approximating the motor parameters $[N/V]$.

3 System Identification

3.1 Direct Measurements

The mass of the cart and the mass of the pendulum were measured directly. The obtained values were:
$m_c = 0.605kg$,
$m_p = 0.088kg$.

3.2 Pendulum Parameters

The length of the pendulum can be determined (assuming that for small angles sin x = x) using the equation:

$$l_p = \frac{T^2 g}{4\pi^2}, \tag{3}$$

where T is the period of free oscillations of the pendulum.

The waveform of several pendulum oscillations was recorded (Fig.1). The average period was read as: $T_{avg} = 1.2489s$. Thus, according to equation (3): $l_p = 0.3876m$.

In order to identify the friction coefficient in the pendulum bearing f_p (with the cart fixed), the following model was assumed:

$$ml^2\ddot{x}_2 + d\dot{x}_2 + mgl\sin x_2 = 0. \tag{4}$$

Assuming $\sin x = x$ equation (4) can be transformed into the form:

$$\ddot{x}_2 + \frac{d}{ml_p^2}\dot{x}_2 + \frac{g}{l_p} = 0. \tag{5}$$

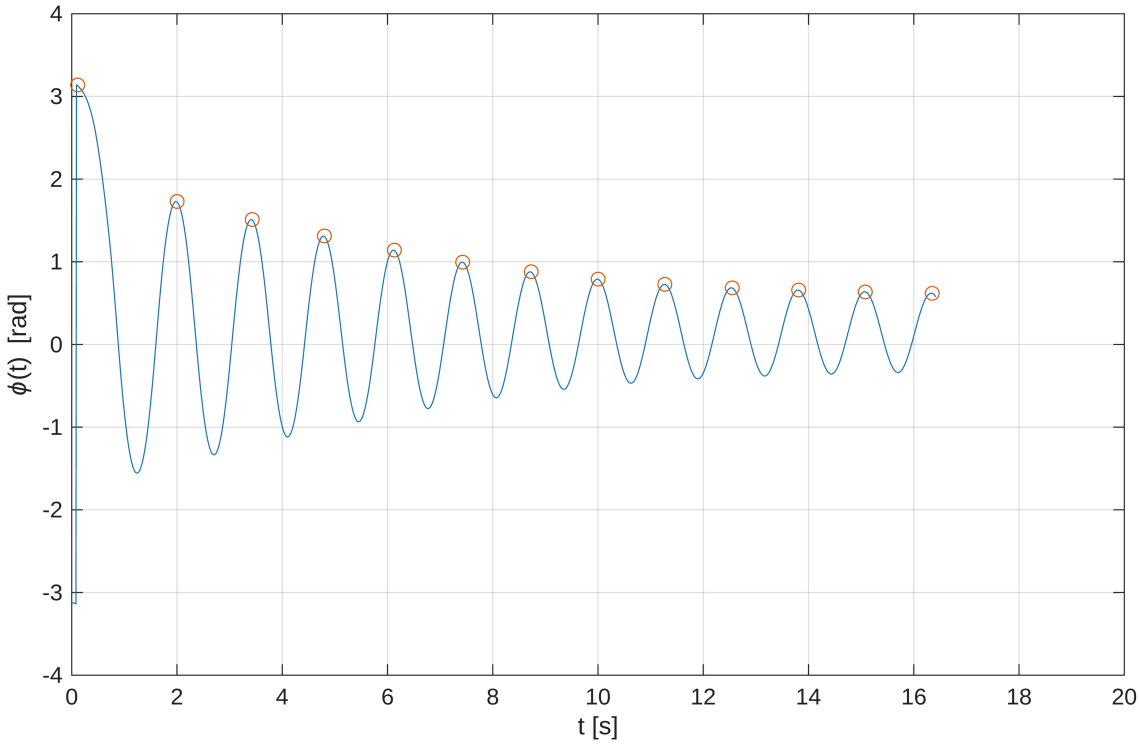

Fig. 1: Pendulum oscillation waveform

Equation (5) was transformed into the state-space representation:

$$\begin{aligned} \dot{x}(t) &= Ax(t), \\ y(t) &= Cx(t), \\ A &= \begin{bmatrix} 0 & 1 \\ -\frac{g}{l_p} & -\frac{f_p}{m_p l_p^2} \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \end{bmatrix}. \end{aligned} \tag{6}$$

The waveform of free oscillations of the pendulum was recorded with the cart fixed. Then, using the least squares method, the value of parameter f_p, was determined for which model (6) produces a similar response. The optimization result is shown in 2. The obtained value: $f_p = 2.6465 \cdot 10^{-4} kg \cdot \frac{m^2}{s}$.

The distance l of the system's center of mass from the pendulum axis can be determined using the equation:

$$l = \frac{m_c l_c + m_p l_p}{m_c + m_p}. \tag{7}$$

where: l_c, l_p distances of the centers of mass of the cart and the pendulum from the pendulum axis, respectively. Assuming $l_c = 0$, we obtain: $l = \frac{m_p l_p}{m_c + m_p} = 0.0492m$.

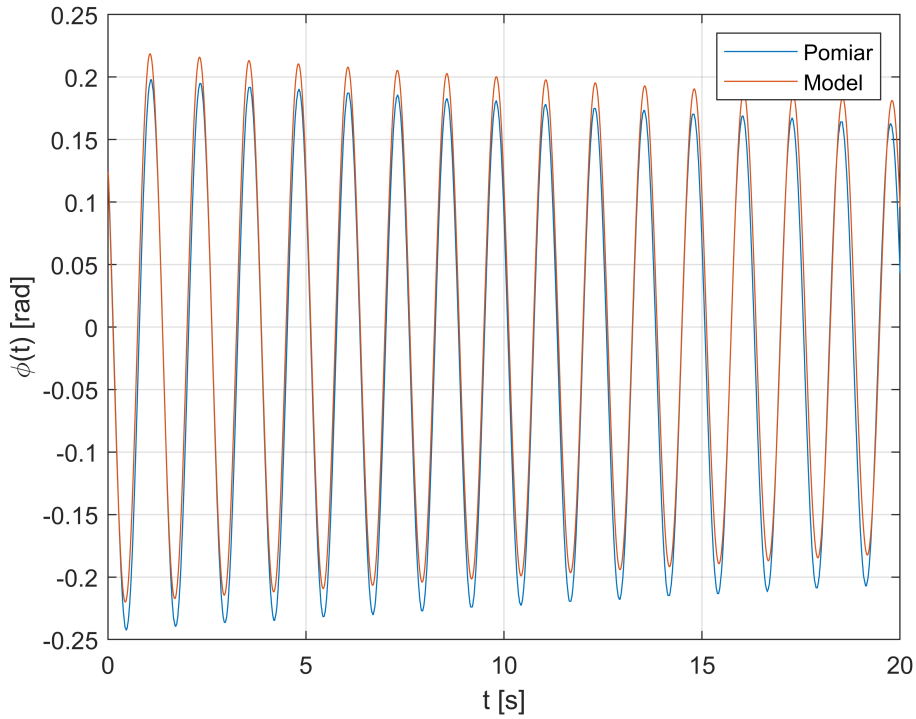

Fig. 2: Result of the optimization of parameter f_p (pendulum friction)

The moment of inertia of the pendulum J_0 was determined directly from the formula for the moment of inertia of a mathematical pendulum:

$$J = mR^2, \tag{8}$$

thus: $J_0 = m_p l_p^2 = 0.0132 kg \cdot m^2$.

3.3 Cart and Motor Parameters

Similarly to the pendulum case, the system consisting of the motor and the cart (with the pendulum removed) can be modeled by the equations:

$$\begin{aligned} F &= m_c \ddot{x}_1 + f_c \dot{x}_1, \\ F &= kU. \end{aligned} \tag{9}$$

After substitution:

$$\ddot{x}_1 + \frac{d}{m_c}\dot{x}_1 - \frac{k}{m_c}U = 0. \tag{10}$$

Equation 10 was transformed into the state-space form:

$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t), \\ y(t) &= Cx(t) + Du(t), \\ A = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{d}{m_c} \end{bmatrix}, B &= \begin{bmatrix} 0 \\ \frac{k}{m_c} \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \end{bmatrix}, D = 0 \end{aligned} \tag{11}$$

The system response to a square wave input was recorded (Fig. 3). Next, similarly to the pendulum case, optimization using the least squares method was performed, this time with respect to two variables: friction coefficient in the cart f_c and coefficient modeling the motor parameters k. The obtained values were: $f_c = 1.88kg/s$, $k = -8.749N/V$. The optimization result is shown in Fig. 4.

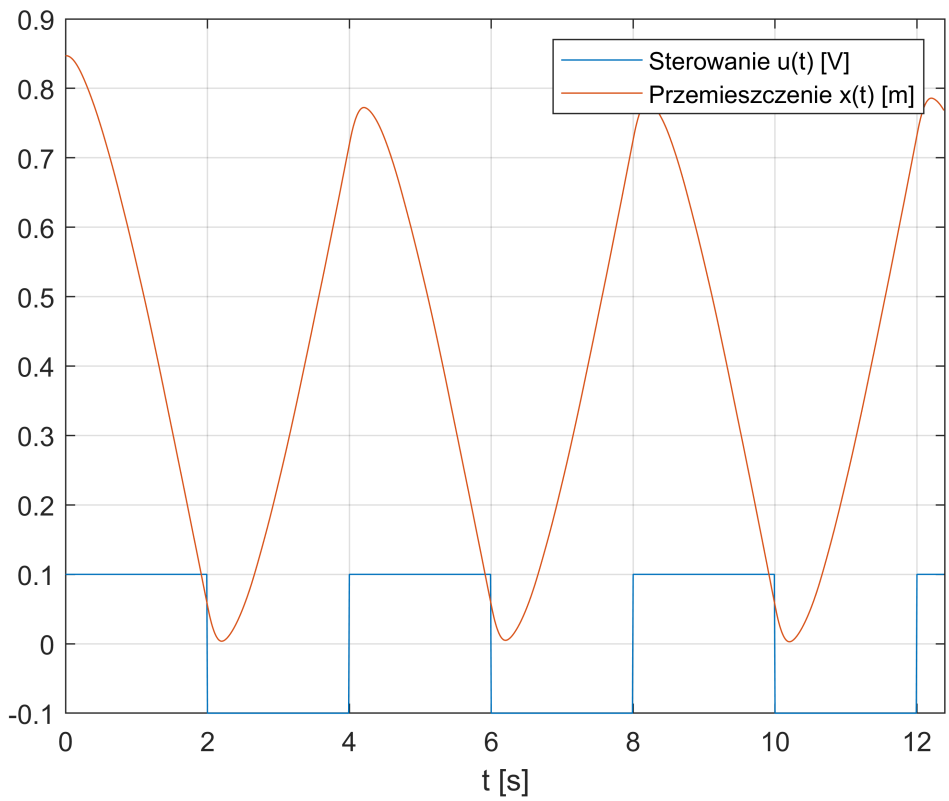

Fig. 3: Measurement of the response of the cart–motor system to square-wave control.

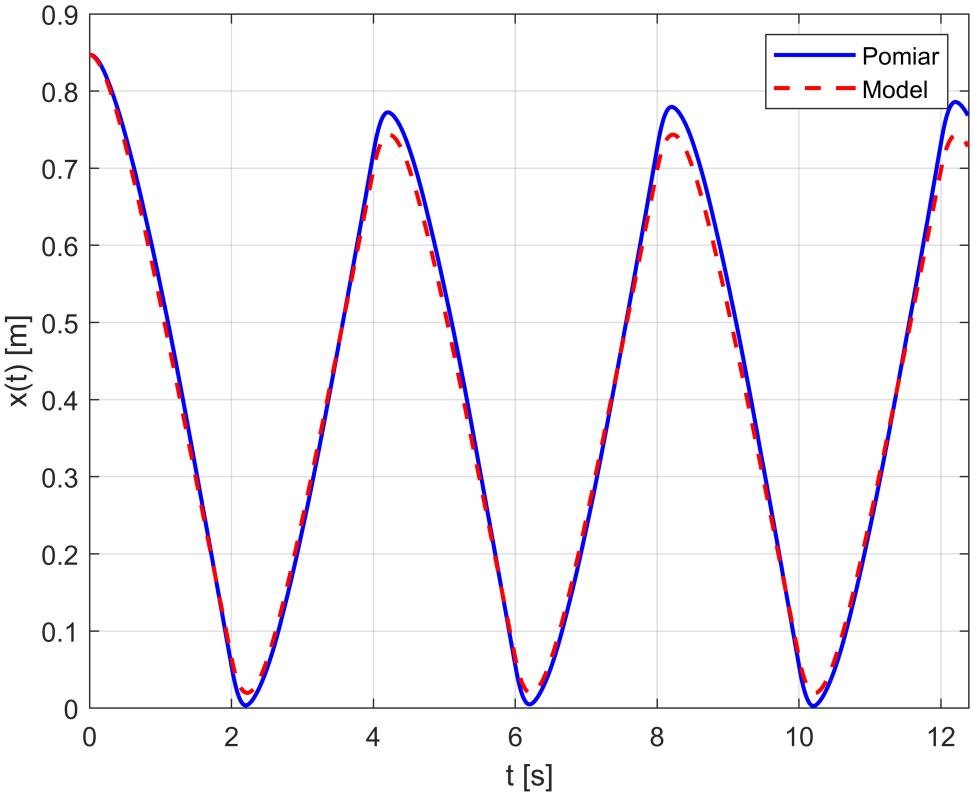

Fig. 4: Result of the cart model optimization

3.4 Summary of Identification Results

Table 1: Determined system parameters

Parameter	Value
m_c	$0.605kg$
m_p	$0.088kg$
f_c	$1.88kg/s$
f_p	$2.6465 \cdot 10^{-4} kg \cdot m^2/s$
l	$0.0492m$
J_0	$0.0132kg \cdot m^2$
k	$-8.7490N/V$

4 Controller Design

4.1 Linearization

The investigated system is strongly nonlinear, therefore in order to design a controller it is first necessary to perform linearization around the operating point.
For a system in the form:

$$\begin{aligned} \dot{x} &= f(x,u), x(0) = x_0, \\ y &= h(x) \end{aligned} \tag{12}$$

we look for a linear approximation in the form:

$$\begin{aligned} \dot{x} &= Ax + Bu, x(0) = x_0, \\ y &= Cx. \end{aligned} \tag{13}$$

For the linearization, the first-order Taylor series approximation was used. Then, in the vicinity of the operating point. (x_r, u_r):

$$\begin{aligned} f(x,u) &= f(x_r,u_r) + \frac{\partial f(x,u)}{\partial x}|_{x=x_r,u=u_r} \cdot (x - x_r) + \frac{\partial f(x,u)}{\partial u}|_{x=x_r,u=u_r} \cdot (u - u_r), \\ h(x) &= h(x_r) + \frac{\partial h(x)}{\partial x}|_{x=x_r,u=u_r} \cdot (x - x_r) \end{aligned} \tag{14}$$

For a multivariable function, we obtain:

$$\begin{aligned} f(x,u) &= f(x_r,u_r) + A(x - x_r) + B(u - u_r), \\ h(x) &= h(x_r) + C(x - x_r), \end{aligned} \tag{15}$$

where A, B, C are the corresponding Jacobian matrices: $A = \frac{\partial f}{\partial x}|_{x_r,u_r}$, $B = \frac{\partial f}{\partial u}|_{x_r,u_r}$, $C = \frac{\partial h}{\partial x}|_{x_r,u_r}$. Additionally, if we assume a reference frame such that $f(x_r,u_r) = 0$, $x_r = 0$ and $u_r = 0$, we obtain a linear approximation of the studied system in the form: (13).

4.2 Controller

In controlling the system, a Linear-Quadratic Regulator (*LQR*) was used. The controller $u = \phi(x)$ minimizes the cost function:

$$J(\phi) = \frac{1}{2}\int_0^\infty (x^T Q x + u^T R u)dt, \tag{16}$$

where Q, R are diagonal weighting matrices assigned respectively to the system's state variables and the control input. If the system is stabilizable or controllable, the optimal regulator for the system $\dot{x} = Ax + Bu$ has the form:

$$u = -Kx \tag{17}$$

The matrix K has the form[1]:

$$K = R^{-1}B^T P, \tag{18}$$

where P is (if it exists) the positive definite solution of the equation:

$$A^T P + PA - PBR^{-1}B^T P + Q = 0. \tag{19}$$

The matrix K was determined using the Matlab command **lqr(A, B, Q, R)**. By trial and error, the following weighting matrices were chosen: $Q = diag(10^{1.5}, 10, 1, 1); R = 1$. The resulting K matrix is::

$$K = [-5.6234, 37.1453, -8.4008, 6.6538] \tag{20}$$

Note: The above controller was calculated based on the linearized model (1),which takes as input the force applied to the cart, not the motor input voltage. The conversion $F = kU$ is directly proportional, so it does not affect the controller's operation; the conversion of the desired force into the voltage supplied to the motor was implemented separately in the control program, independently of the controller.

5 Simulation

Before implementing the controller on the real system, a simulation in Simulink was performed. A model was created as shown in Fig. 5.
The simulation result is shown in Fig. 6.

6 Test on the Real System

A practical test of the controller was conducted, and the test was successful. The recorded state variable trajectories are shown in Fig. (7). The cart displacement and pendulum angle are measured relative to the position where the cart is at the center of the track and the pendulum is in the upright position.

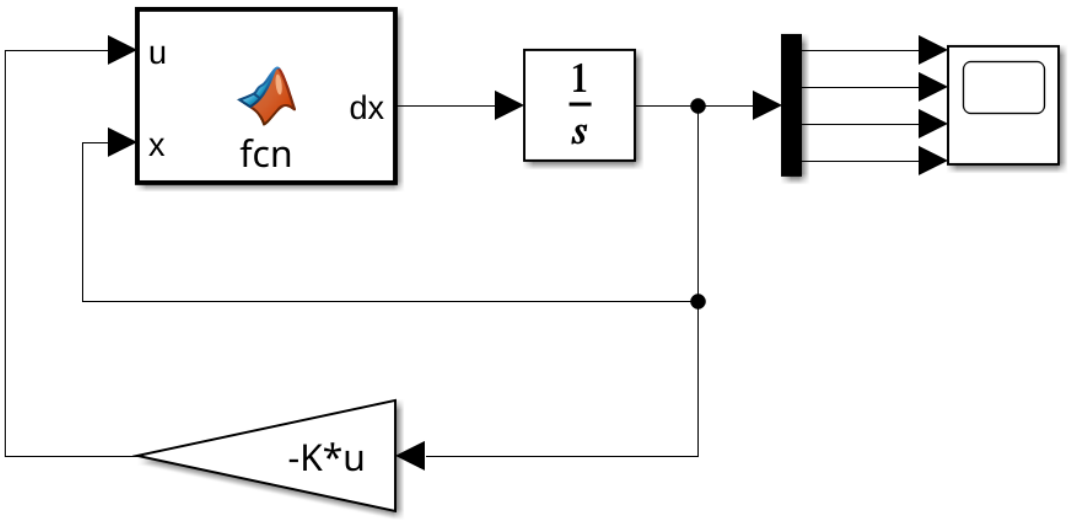

Fig. 5: Simulink Model of the System

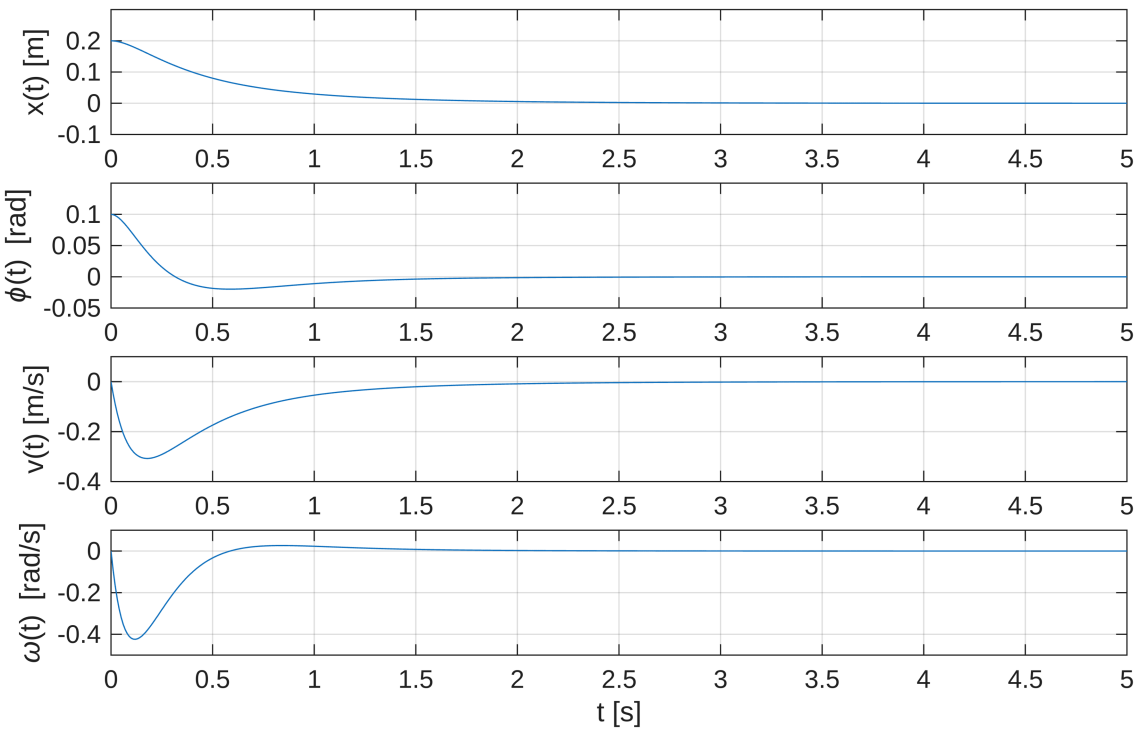

Fig. 6: Simulation of the system response with the LQ controller to a small deviation from the equilibrium point

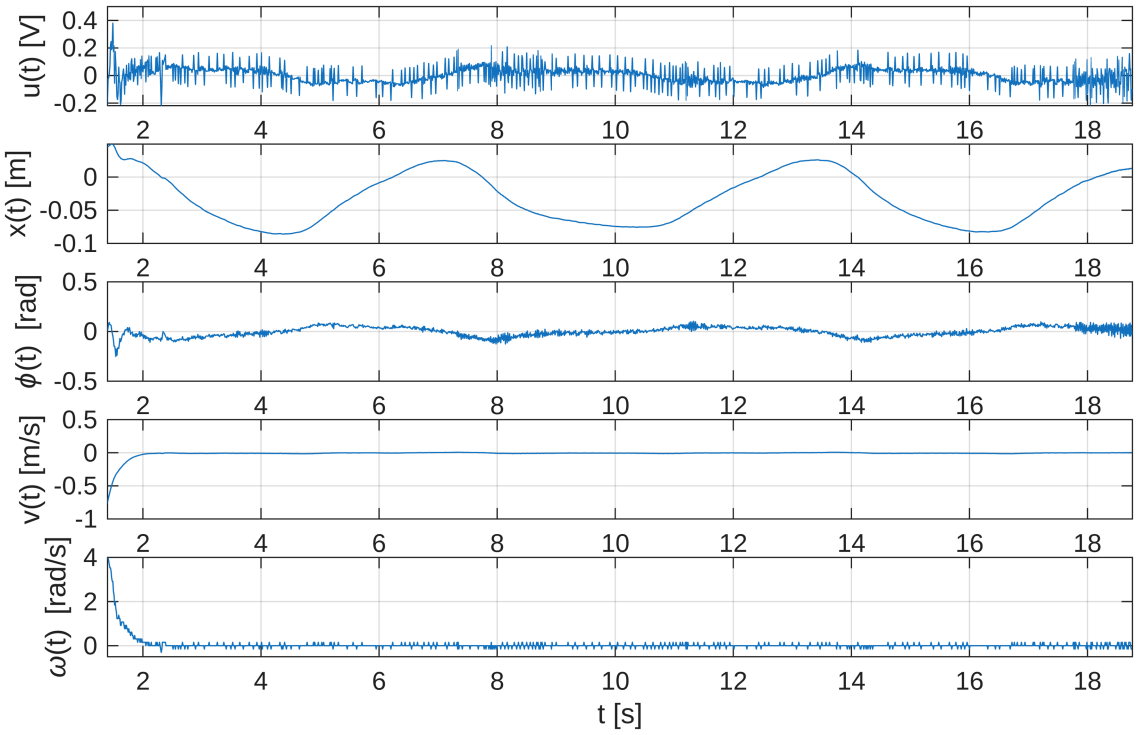

Fig. 7: Recorded trajectories from the practical test of the controller

Literatura

[1] K. Ogata, *Modern Control Engineering*, Prentice Hall, 2010.